Ahmedabad University

CSE 631: MachineLearning

Weekly Report(7)

(Group - 9) Learning Machine

Section 1

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# Weekly Progress Report - 7

## Machine Learning–Based Automated Stone Counting System for Embroidered Fabrics

## 1. Things that have been accomplished this week.

The ongoing stage of the project was aimed at the replacement of the classical methods of machine learning with the deep learning methods of enhanced accuracy and strength in detecting stones.

### 1.1 Transition from HOG to CNN

- According to the past experience, the HOG feature extraction was eliminated because of its weakness to address complicated and dense stone designs.
- Automatic learning of features with Introduced Convolutional Neural Networks (CNN).
- CNN enabled:
    - Enhanced feature of space.
    - Determined and overlapping stone patterns are better identified.
    - Less reliance on manual features engineering.

### 1.2 Image Preprocessing Enhancements

- Further CLAHE (Contrast Limited Adaptive Histogram Equalization) use to enhance contrast.
- Standardization of input images was done by use of normalization techniques.
- Resizing of images was introduced to fit CNN feed.
- These were to enhance stability and convergence of models as they were trained.

### 1.3 CNN Model Development

- Developed and trained a CNN-based model of classification/detection:
    - Various convolutional layers of feature extraction.
    - Dimensionality reduction through pooling.
    - Classification Fully connected layers.
- ReLU activation functions were taken.
- Sigmoid/Final output prediction with Applied Softmax.

### 1.4 Training and Validation Strategy

- Database divided into training and validation.
- Generalization was done using applied 5-fold cross-validation.
- Used:
    - Loss functions (Binary Cross-Entropy / Categorical Cross-Entropy).
    - Optimization algorithms (Adam optimizer)
- Training observed in terms of accuracy and loss.

### 1.5 Sliding Window Detection with CNN

- Reused sliding window plan incorporated with CNN:
    - All the windows were taken through CNN to be classified.
- Tuned:
    - Window size
    - Stride
- This enhanced accuracy of the detection with respect to HOG-based pipeline.

### 1.6 CNN Based Tile-Based Detection

- Further adoption of tile based method of large images:
    - Pictures subdivided into little squares.
    - CNN operated on individual tiles.
- Benefits observed:
    - Enhanced management of the thick territories.
    - Higher accuracy in detection in high concentration areas of stones.

### 1.7 Hard Negative Mining (Improved)

- Hard negative mining continued with CNN:
    - False positives gathered due to CNN predictions.
    - Training of a new model on these hard samples.
- Result:
    - Considerable improvement in false arrests.
    - Enhanced strengths of the CNN model.

### 1.8 Detection Merging and Post-processing

- Better inter tile merging of detections:
    - Less overlapping bounding boxes.
    - The Applied Non-Maximum Suppression (NMS).
- Increased visualization of the final output with more distinct bounding boxes.

### 1.9 Model Evaluation

- Measures of model performance using:
    - Accuracy
    - Precision
    - Recall
    - F1-score

Observations:

- CNN greatly performed better than HOG + SVM pipeline.
- Better localization of complicated and dense embroidered designs.
- Improved learning in different lighting and texture.

### 1.10 Key Findings

- The use of CNN-based approach removed the constraints in the manual extraction of features.
- CLAHE was still having a positive effect on performance.
- The tile-based detection with CNN yielded good results.
- Hard negative mining was also used to increase the robustness of the models.
- Deep learning method was more applicable to the real-life application.

## 2. Activities to be done within the next week.

- Tune CNN architecture in terms of improved performance and less computation.
- Test more complicated architectures (e.g. deeper CNN or transfer learning models).
- Enhance the level of detection speed to make it real-time.
- Manually adjust the hyperparameters (learning rate, batch size, epochs).
- Improves the localization of the bounding box and accuracy.
- Continue working on deployment pipeline (integration with application/system).
- Further improve metrics of evaluation (precision, recall, F1-score).
- Improve the performance by trying the object detection models such as YOLO or Faster R-CNN.